

CryptoVision AI

AI-Powered Cryptocurrency Forecasting & Analytics Platform

Project Summary & Integration Report

Project Overview

CryptoVision AI is a full-stack, AI-driven web application designed to analyze cryptocurrency market data, generate predictive insights, and provide interactive visualization tools for traders and learners.

Objectives

- ▶ Build a real-time crypto analytics dashboard
- ▶ Apply machine learning models for price prediction
- ▶ Provide interactive technical indicators (MA, RSI)
- ▶ Enable user-driven scenario simulation
- ▶ Create a clean, professional, trading-style UI

System Architecture



Frontend

Next.js (App Router) with TypeScript, Tailwind CSS, and ShadCN UI. Real-time data visualization via Lightweight Charts.



Backend

FastAPI (Python) based RESTful architecture. Efficient model serving and API endpoint management.



Machine Learning

XGBoost & Random Forest models for predicting market directions (BUY/SELL/HOLD) and expected price movements based on historical OHLC data.

Core Features



Live Dashboard

Real-time price streams with auto-refresh capabilities for up-to-the-second market analysis.



Interactive Charts

Advanced Candlestick visualization with MA(7) and MA(21) overlays.



Crosshair Panel

Precision data lookup for price, indicators, and volume via time-based mapping.



AI Predictions

Market trend analysis and confidence scores provided by integrated ML inference.

Technical Solutions

Crosshair Data: Solved update lag by replacing index-based mapping with **time-based lookup** using timestamp matching.

Indicators: Solved "NaN" values for MA/RSI by extracting data directly from **computed calculation arrays** rather than chart API limitations.

UI/UX Stability: Implemented **fixed container heights** and overflow management to eliminate chart clipping on responsive layouts.

Project Structure

```
crypto-project/  
├── backend/  
│   ├── main.py (FastAPI)  
│   ├── predictor.py (Inference Logic)  
│   └── model_loader.py (Weights & Serialization)  
├── frontend/  
│   ├── app/ (Next.js Routes)  
│   └── components/ (UI & Charts)
```

UI/UX Highlights

- ▶ Dark theme fintech layout
- ▶ Modular component-based design
- ▶ Responsive, clean animations

Deployment Plan

Frontend: Vercel • **Backend:** Render

Future Enhancements

- ▶ ☐ MACD & Bollinger Band indicators
- ▶ ☐ Real-time WebSocket streaming
- ▶ ☐ Multi-timeframe support (1H, 4H, 1D)
- ▶ ☐ User authentication & Portfolio tracking

Conclusion

CryptoVision AI demonstrates a seamless synthesis of performance-driven full-stack development and predictive machine learning.

Developed for Analytical Clarity & Educational Insight